

## **FITNESS-TRACKER-ANALYTICS-DASHBOARD | Power BI**

Data analysis and visualization project completed during DevAlpha internship

### **PROJECT OVERVIEW**

This project was developed as part of my DevAlpha Internship (Task 2: Visualization Study). The objective was to transform raw fitness tracker data into meaningful visual insights using Power BI and present findings professionally.

The dashboard provides a comprehensive analysis of user fitness behavior, activity levels, calories burned, sleep patterns, heart rate trends, and exercise distribution.

### **TOOLS USED**

Microsoft Power BI

Microsoft Excel

DAX (Data Analysis Expressions)

### **BUSINESS PROBLEM**

Fitness tracking devices generate large volumes of health and activity data. However, raw data alone does not provide actionable insights. This project aims to visualize key performance indicators and behavioral trends that support informed fitness and wellness decisions.

### **KEY KPIs**

KPI      Value

Average Steps 7.89K

Average Calories Burned      330 kcal

Average Sleep Duration      7.4 Hours

Average Heart Rate      121 BPM

Total Calories Burned 100,201 kcal

Total Steps      2 million+

Dashboard Features

Activity Analysis

Average Steps by Activity Level

Exercise Type Distribution

Average Steps Trend by Date

Health & Wellness Analysis

Average Heart Rate by Activity Level

Sleep Duration Monitoring

Performance Analysis

Calories Burned Tracking

Exercise Frequency Distribution

## **KEY INSIGHTS**

### 1. Cardio Dominates User Activity

Cardio exercises account for the highest proportion of recorded activities.

### 2. Moderate Activity Users Form the Largest Segment

Over 50% of users fall within the Moderate Activity category.

### 3. Higher Activity Leads to Greater Energy Expenditure

Users with higher activity levels consistently burn more calories.

### 4. Healthy Sleep Patterns

Average sleep duration of 7.4 hours aligns with recommended sleep guidelines.

### 5. Heart Rate Increases with Activity Intensity

Users in the High Activity group recorded the highest average heart rates.

## **Recommendations**

1. Implement personalized step goals.

2. Introduce activity streaks and rewards.

3. Integrate sleep quality monitoring.

4. Provide adaptive fitness recommendations based on activity behavior.

5. Use heart-rate insights to optimize training plans.

6. Dashboard Preview

